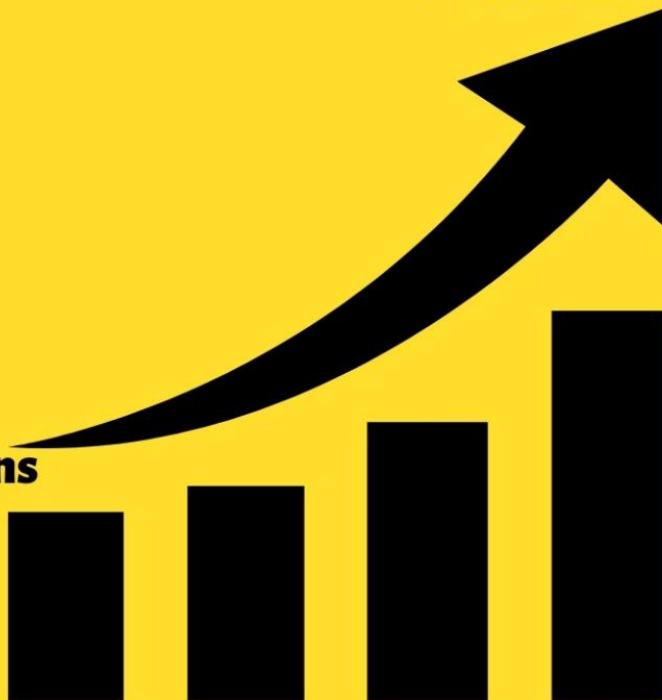


TIME & DISTANCE

ALL TYPE WISE QUESTIONS

Advance Level Questions

Basic Questions



Maths with Gajanand



Time and Distance

$$\text{Distance} = \text{Time} \times \text{Speed}$$

$$D = S \times T$$

$$1 \text{ km/Hr} = \frac{1 \text{ km}}{1 \text{ Hr}} = \frac{1000 \text{ m}}{60 \times 60 \text{ sec}} = \frac{5}{18} \text{ m/sec}$$

$$1 \text{ km/Hr} = \frac{5}{18} \text{ m/sec}$$

$$1 \text{ m/sec} = \frac{18}{5} \text{ km/Hr}$$

Case-I : if Distance = Constant $S \propto \frac{1}{T}$, $T \propto \frac{1}{S}$

$$S = x : y$$

$$T = y : x$$

$$\frac{S_1}{S_2} = \frac{T_2}{T_1}$$

Case-II :- If Time = Constant

$$D \propto S$$

$$S = x : y$$

$$D = x : y$$

$$\frac{D_1}{D_2} = \frac{S_1}{S_2}$$

Case-III : If Speed = Constant $D \propto T$

$$D = x : y$$

$$T = x : y$$

$$\frac{D_1}{D_2} = \frac{T_1}{T_2}$$

- 1 Rakesh walking at the speed of 8 km/hr crosses a bridge in 30 minutes. What is the length of the bridge in kilometres?

Soln:-

$$\begin{aligned} \textcircled{1} &= S \times T \\ &= 8 \times \frac{30}{60} \\ &= \boxed{4 \text{ km}} \text{ Ans} \end{aligned}$$

- 2 A car covered 150 km in 5 hours. If it travels at one-third its usual speed, then how much more time will it take to cover the same distance?

Soln:-

$$\begin{aligned} \text{initial speed} &= \frac{150}{5} = 30 \text{ km/h} \\ \text{New speed} &= \frac{30}{3} = 10 \text{ km/h} \\ \text{Time} &= \frac{150}{10} = 15 \text{ hr} \\ \text{time difference} &= 15 - 5 \\ &= \boxed{10 \text{ hour}} \text{ Ans} \end{aligned}$$

- 3 A car travelling at an average speed of 72 km/hr takes 9 minutes to travel a certain distance. By how much should it increase its speed (in km/hr) to travel the same distance in 8 minutes?

Soln:-

$$\begin{aligned} \text{Distance} &= 72 \times \frac{9}{60} \\ &= \frac{54}{5} \text{ km} \\ \text{New speed} &= \frac{\frac{54}{5}}{\frac{8}{60}} = \frac{54}{5} \times \frac{60}{8} \\ &= 81 \text{ km/hr} \end{aligned}$$

m-II

$$\begin{aligned} \text{New speed} &= \frac{72 \times 9}{8} \\ &= 81 \text{ km/h} \\ \text{change} &= 81 - 72 \\ &= \boxed{9 \text{ km/h}} \text{ Ans} \end{aligned}$$

$$\text{Increase speed} = 81 - 72 = \boxed{9 \text{ km/hr}} \text{ Ans}$$

- [4] The distance between two cities is covered in $3\frac{1}{4}$ hours at a speed of 52 km/hr. If the speed is increased to 65 km/hr. how much time would be saved?

Soln:-

$$D = 52 \times \frac{13}{4}$$

$$D = 169 \text{ km}$$

$$\text{New Time} = \frac{169}{65} = \frac{13}{5} \text{ hr}$$

$$\text{Time saved} = \left(\frac{13}{4} - \frac{13}{5} \right) \times 60 = \boxed{39 \text{ min}} \text{ Ans}$$

- [5] A bullet fired from a rifle travels at an average speed of 2520 km/hr. It hits the target after 0.2 seconds. How far (in m) is the target from the rifle?

Soln:-

$$D = \frac{2520 \times 0.2 \times 1000}{60 \times 60}$$

$$= \boxed{140 \text{ meters}} \text{ Ans}$$

- [6] A train covers a distance in 30 minutes. If it runs at a speed of 54 km/hr on an average. The speed at which the train must run to reduce the time of the journey to 20 minutes is:

Soln:-

$$D = 54 \times \frac{30}{60}$$

$$= 27 \text{ km}$$

$$\text{New speed} = \frac{27 \times 60}{20}$$

$$= \boxed{81 \text{ km}} \text{ Ans}$$

- [7] A train covers a distance of 12 km in 12 minutes. If its speed is decreased by 5 km/hr, then the time taken by it to cover the distance of 22 km will be?

Soln:-

$$\text{Speed} = \frac{12 \text{ km}}{12} \times 60 = 60 \text{ km/hr}$$

$$\text{New speed} = 60 - 5 = 55 \text{ km/hr}$$

$$\text{New Time} = \frac{22}{55} \times 60 = \boxed{24 \text{ minutes}} \quad \underline{\text{Ans}}$$

- [8] A man can cover a certain distance in 3 hours 36 minutes if he walks at the rate of 5 km/hr. If he covers the same distance on cycle at the rate of 24 km/hr, then the time taken by him in minutes is?

Soln:-

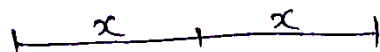
$$\text{Distance} = 5 \times 3 \frac{36}{60} = 5 \times 3 \frac{3}{5}$$

$$= 5 \times \frac{18}{5} = 18 \text{ km}$$

$$\begin{aligned} \text{time} &= \frac{18}{24} \times 60 \\ &= \boxed{45 \text{ min}} \quad \underline{\text{Ans}} \end{aligned}$$

- [9] Mohan finishes a journey by scooter in 5 hours. He travels the first half of the journey at 30 km/h and the second half of the journey at 20 km/h. The distance covered by him is:

Soln:-



$$\frac{x}{30} + \frac{x}{20} = 5$$

$$\frac{2x + 3x}{60} = 5$$

$$x = 60$$

$$\text{Total Distance} = \boxed{120 \text{ km}} \quad \underline{\text{Ans}}$$

fm-ii

$$\text{avg speed} = \frac{2xy}{x+y}$$

$$= \frac{2 \times 30 \times 20}{30 + 20}$$

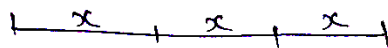
$$= 24 \text{ km/hr}$$

$$\text{D} = 24 \times 5$$

$$= \boxed{120 \text{ km}} \quad \underline{\text{Ans}}$$

10 Mohan travels three equal distance at speeds of 12 km/hr , 18 km/h , and 24 km/h . If he takes a total of 13 hours , then what is the total distance covered?

Soln:-



$$\frac{x}{12} + \frac{x}{18} + \frac{x}{24} = 13$$

$$\frac{6x + 4x + 3x}{72} = 13$$

$$x = 72$$

$$\begin{aligned}\text{Total Distance} &= 72 \times 3 \\ &= \boxed{216} \text{ km} \quad \underline{\text{Ans}}\end{aligned}$$

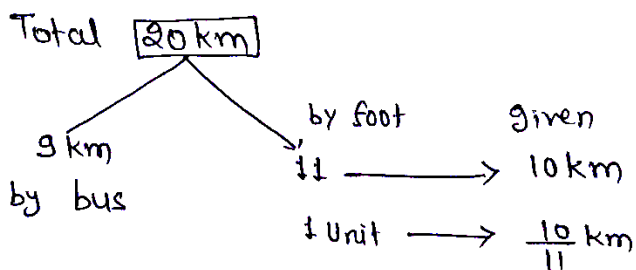
11] A bullock cart has to cover a distance of 120 km in 15 hours. If it covers half of the journey in $\frac{3}{5}$ th time, the speed to cover the remaining distance in the time left has to be ?

Soln:-

Speed = $\frac{60}{6}$ ← Remaining Distance = $120 - 60$
 = $\boxed{10 \text{ km/hr}}$ Ans ← Remaining Time = $15 \times \frac{2}{5}$
 = 6 hr

[12] A man covers $\frac{9}{20}$ distance by bus and the remaining 10 km on foot. His total journey (in km) is:

Soln. -



Total 20 Unit $\rightarrow \frac{10}{11} \times 20$
 $= \boxed{18.18 \text{ km}}$ Ans

- [13] Suresh goes on a trip on his motor-cycle and rides for 410 km. If he rides for 5 hours at a speed of 50 km/hr, find at what speed he travels for the remaining 4 hours of the journey?

Soln:-

$$\text{Total Distance} = 410$$

$$\text{Covered Distance} = 50 \text{ km/hr} \times 5 \text{ hr} = 250$$

$$\begin{aligned} \text{Remaining} &= 410 - 250 \\ &= 160 \text{ km} \end{aligned}$$

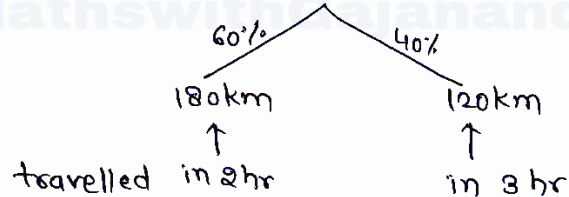
$$\text{Speed} = \frac{160}{4} = \boxed{40 \text{ km/hr}} \quad \underline{\text{Ans}}$$

- [14] Travelling at 60 km/hr, a person reaches his destination in a certain time. He covers 60% of his journey in $\frac{2}{5}$ th of the time. At what speed (in km/h) should he travel to cover the remaining journey so that he reaches the destination right on time?

Soln:-

$$\text{Let the total time} = 5 \text{ hours}$$

$$\begin{aligned} \text{Distance covered in 5 hours} &= 60 \times 5 \\ &= 300 \text{ km} \end{aligned}$$



$$\text{Speed} = \frac{120}{3} = \boxed{40 \text{ km/hr}} \quad \underline{\text{Ans}}$$

- [15] A person travels $5x$ distance at a speed of 5 km/h; x distance at a speed of 5 km/hr and $4x$ distance at a speed of 6 km/hr and takes a total of 112 minutes. What is the total distance travelled by the person?

Soln:-

$$\frac{5x}{5} + \frac{x}{5} + \frac{4x}{6} = \frac{112}{60}$$

$$x = 1$$

$$\text{Total Distance} = 5x + x + 4x = 10x = \boxed{10 \text{ km}}$$

16 A person has to cover a distance of 160km in 15 hours. If he covers $\frac{4}{5}$ of the distance in $\frac{2}{3}$ of the time, then what should be his speed (in km/hr) to cover the remaining distance in the remaining time?

Soln:- Remaining Distance = $160 \times \frac{1}{5} = 32 \text{ km}$

Remaining Time = $15 \times \frac{1}{3} = 5 \text{ hr}$

Speed = $\frac{32}{5} = \boxed{6.4 \text{ km/hr}}$ Ans

17 A man walk at a speed of 8km/h. After every kilometer he takes a rest of 4 minutes. How much time will he take to cover a distance of 6 km?

Soln:- Time = $\frac{6}{8} = \frac{3}{4} \text{ hour}$
= 45 min.

Including rest = $45 + 5 \times 4 \text{ min}$
= $\boxed{65 \text{ min}}$ Ans.

18 A bus starts running with some initial speed and its speed increased every hour by 9km/hr. If it takes 11 hours to cover a distance of 572km, then what was the initial speed (in km/hr) of the bus.

Soln:- Let initial speed = a

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$572 = \frac{11}{2} [2a + (11-1) \times 9]$$

$$52 \times 2 = 2a + 90$$

$$a = \boxed{7 \text{ km/hr}}$$
 Ans

19 A bus start running with the initial speed of 21 km/hr and its speed increased every hour by 3 km/hr. How a many hours will its take to cover a distance of 252 km?

Soln:-

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$d = 3 \text{ km/hr}$$

$$a = 21 \text{ km/hr}$$

$$252 = \frac{n}{2} [2 \times 21 + (n-1)3]$$

$$504 = 3n^2 + 39n$$

$$= 3n^2 + 39n - 504 = 0$$

Solve $\boxed{n=8}$, $n=-21$
 $\quad \quad \quad \underline{\underline{\text{Ans}}}$ $\quad \quad \quad \times$

20 A bus starts running with the initial speed of 33 km/hr and its speed increased every hour by certain amount. If it takes 7 hours to cover a distance of 315 km, then what will be hourly increment (in km/hr) in the speed of the bus? [SSC CHSL - 2015]

Soln:-

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$n = 7$$

$$a = 33$$

$$315 = \frac{7}{2} [2 \times 33 + (7-1)d]$$

$$315 = \frac{7}{2} [66 + 6d]$$

$$90 - 66 = 6d$$

$$d = \frac{24}{6}$$

$$d = \boxed{4 \text{ km/hr}} \quad \underline{\underline{\text{Ans}}}$$

[21] Two cars start from the same place at the same time at right angles to each other. Their speeds are 54 km/hr and 72 km/hr, respectively. After 20 seconds the distance between them will be:

Soln:-

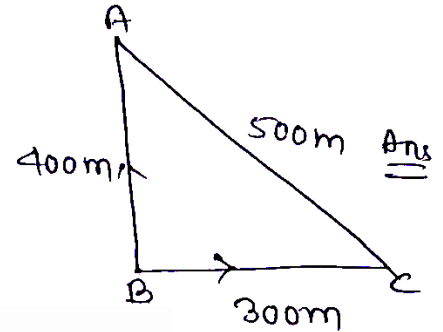
$$54 \text{ km/hr} \times \frac{5}{18} = 15 \text{ m/sec}$$

$$\text{The Distance of BC} = 15 \times 20 \\ = 300 \text{ m}$$

$$72 \text{ km/hr} \times \frac{5}{18} = 20 \text{ m/sec}$$

$$\text{The Distance of AB} = 20 \times 20 \\ = 400 \text{ m}$$

$$AC = \sqrt{(300)^2 + (400)^2} = \boxed{500 \text{ m}} \text{ Ans}$$



[22] A car moves a distance of 600 km with uniform speed.

The number of hours taken for the journey is $\frac{2}{3}$ of the number representing speed in km/hr. The time taken to cover the distance is:

Soln:-

$$T : S = 2x : 3x$$

$$\text{Distance} = 2x \times 3x$$

$$600 = 6x^2$$

$$x = 10$$

$$\text{Time} = 2x$$

$$= \boxed{20 \text{ hour}} \text{ Ans}$$

Q3 A distance is covered by a cyclist at a certain speed. If a jogger covers half the distance in double the time, the ratio of the speed of the jogger to that of the cyclist is:

Soln:-

	Jogger	:	Cyclist
Distance	1	:	2
time	2	:	1

$$\text{Ratio of speed} = \frac{1}{2} : \frac{2}{1} \Rightarrow \boxed{1:4} \text{ Ans}$$

Q4 If a distance of 50m is covered in 1 minute, 90m in 2 minutes and 130m in 3 minutes. Find the distance covered in 15th minute.

Soln:-

50m, 90m, 130m ----- A.P

$$\begin{aligned} T_n &= a + (n-1)d \\ &= 50 + (15-1) \times 40 \\ &= \boxed{610m} \text{ Ans} \end{aligned}$$

Q5 After excluding stoppages, the speed of a bus is 60km/hr and after including stoppages, it is 45km/hr. For how many minutes does the bus stop per hour? [SSC CHSL-2016]

Soln:- No. of minutes bus per hour

$$= \frac{\text{excluding stoppage} - \text{including stoppage}}{\text{excluding stoppage}} \times 60$$

$$= \frac{60 - 45}{60} \times 60$$

$$= \boxed{15 \text{ minute}} \text{ Ans}$$

Q6] The speed of train A is 25 km/hr more than the speed of train B. A takes 4 hours less time to travel a distance of 300 km than what train B takes to travel 250 km. What is the speed (in km/hr) of A?

(a) 60

Soln:-

$$\text{Speed of A} = (x + 25) \text{ km/hr}$$

(b) 50

$$\text{Speed of B} = x \text{ km/hr}$$

(c) 65

$$\frac{250}{x} - \frac{300}{x+25} = 4$$

(d) 55

Now from option (b)

$$A = \boxed{50 \text{ km/hr}}$$

$$B = 25 \text{ km/hr} \quad \text{satisfying } \checkmark$$

Q7] The diameter of each wheel of car is 70 cm. If each wheel rotates 400 times per minute, then the speed of the car (in km/hr) [SSC CHSL-2017]

Soln:-

$$\begin{aligned} \text{Circumference of wheel} &= 2\pi r \\ &= 2 \times \frac{22}{7} \times \frac{70}{2} = 220 \text{ cm} \end{aligned}$$

$$\text{Speed per hour} = \frac{220 \times 400 \times 60}{100 \times 1000} = \boxed{52.8 \text{ km/h}}$$

Q8] The wheel of a car has 210 cm diameter. How many revolutions per minute must the wheel make so that the speed of the car is kept at 120 km/h?

$$\text{Soln:-} \quad \text{Circumference} = 2 \times \frac{22}{7} \times \frac{210}{2} = 660 \text{ cm} = 6.6 \text{ m}$$

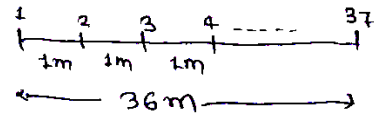
$$6.6 \times n = \frac{120 \times 1000 \text{ m}}{60} \quad \leftarrow \text{Covered distance in 1 minute}$$

$$n = \frac{2000}{6.6} = \boxed{303.03} \text{ Ans}$$

29 37 trees are planted in straight line such that distance between any two consecutive trees is same. A car takes 20 seconds to reach the 13th tree. how much more time (in sec) will take to reach the last tree?

Soln:-

Let Total distance = 36m



$$\begin{array}{l} 12m \longrightarrow 20 \text{ seconds} \\ \times 3 \left(\begin{array}{l} 36m \longrightarrow \end{array} \right) \times 3 \end{array}$$

60 seconds Ans

30 29 boys are standing in a line such that distance between any two consecutive boys is same. A teacher takes 7 sec. to reach from first boy to 15th boy. How much more time (in sec) will he take to reach the 22nd boy?

Soln:-

Total distance = 28m

$$\begin{array}{l} 14m \longrightarrow 7 \text{ sec} \\ 1m \longrightarrow 0.5 \text{ sec} \end{array}$$

then $21m \longrightarrow 21 \times 0.5 = 10.5 \text{ sec}$

more time taken = $10.5 - 7 = \boxed{3.5 \text{ sec.}}$ Ans

31 Rahul and Mithun travel a distance of 30 km. The sum of their speeds is 70 km/h and the total time taken by both to travel the distance is 2 hours 6 minutes. The difference between their speeds is:

Soln:-

$$\frac{30}{x} + \frac{30}{y} = \frac{21}{10}$$

$$x + y = 70 \quad \text{--- (1)}$$

$$2 \text{ hr, } 6 \text{ min} = \frac{120 + 6}{60} = \frac{21}{10}$$

$$30 \left(\frac{x+y}{xy} \right) = \frac{21}{10}$$

$$30 \times \frac{70}{xy} = \frac{21}{10}$$

$$xy = 1000 \quad \text{--- (2)}$$

$$(x-y)^2 = (x+y)^2 - 4xy$$

$$(x-y)^2 = 4900 - 4000$$

$$(x-y)^2 = 900$$

$$x-y = \boxed{30 \text{ km/hr}} \text{ Ans.}$$

[32] X and Y travel a distance of 90 km each such that the speed of Y is greater than that of X. The sum of their speeds is 100 km/hr and the total time taken by both is 3 hours 45 minutes. The ratio of the speed of X to that of Y is:

[SSC CHL Mains-2020]

Soln $X + Y = 100 \text{ km/hr} \text{ — ①}$

$$\frac{90}{X} + \frac{90}{Y} = \frac{15}{4}$$

$$90 \left(\frac{x+y}{xy} \right) = \frac{15}{4}$$

$$90 \times \frac{100}{xy} = \frac{15}{4}$$

$$xy = 2400 \text{ — ②}$$

from
① 2 ②

$$x = 40, y = 60 \text{ [Y > X]}$$

$$40 : 60 = \boxed{2 : 3} \text{ Ans}$$

[33] A car driver leaves Bangalore at 8.30 a.m. and expected to reach a place 300 km away from Bangalore at 12.30 p.m. At 10:30 he finds that he has covered only 40% of the distance. By how much he has to increase the speed of the car in order to keep up his schedule?

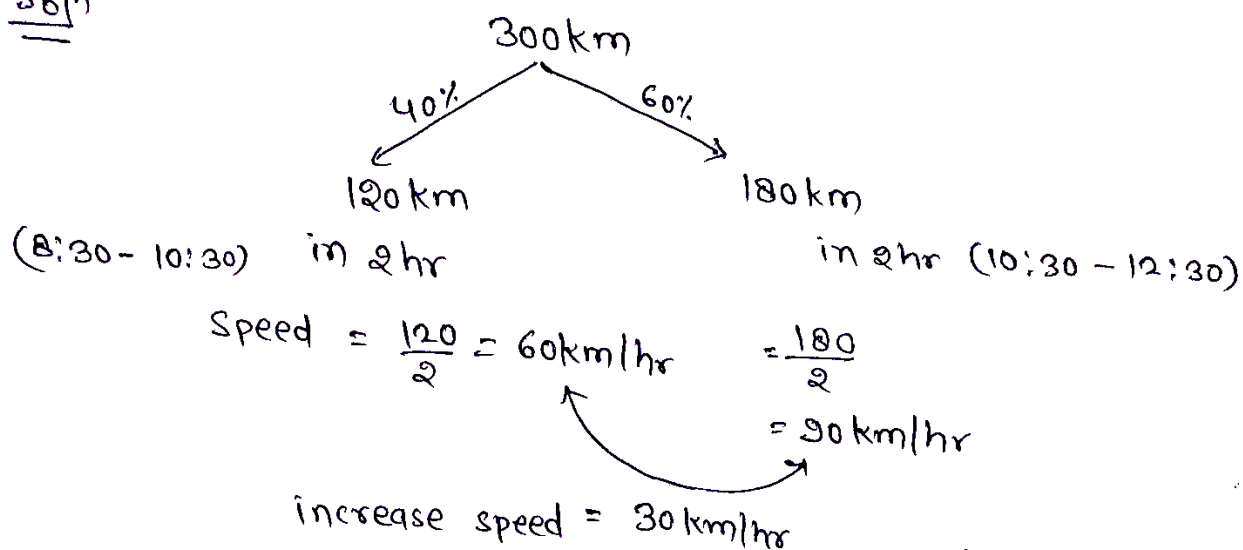
(a) 45 km/hr

(b) 40 km/hr

(c) 35 km/hr

(d) 30 km/hr

Soln



Concept

IF Distance is constant

$$S \propto \frac{1}{T}$$

$$S :- x : y$$

$$T :- y : x$$

34 The speed of a train is 220% of the speed of a car. The covers a distance of 950 km in 19 hours. How much distance will the train cover in $3\frac{1}{2}$ hours?

Soln:-

$S \Rightarrow$	Train	:	Car
	11	:	5
	$\downarrow \times 10$		$\downarrow \times 10$
	110 km/hr		50 km/hr

Speed of car

$$\frac{950}{19} = 50 \text{ km/hr}$$

$$\text{Distance} = 110 \times 3\frac{1}{2} \text{ hour}$$

$$= \boxed{385 \text{ km}} \text{ Ans}$$

[35] The speed of two trains are in the ratio 6:7. If the second train runs 364 km in 4 hours, then the speed of first train is:

Soln:-

I		II
Speed:	6	7
	:	:
	x13	x13
	↓	↓
	78 km/hr	91 km/hr

Ans

$$S_2 = \frac{364}{4} = 91 \text{ km/hr}$$

[36] A bus covers a 60 kilometre distance in 1 hour 30 minutes whereas the same distance is covered by a car in 45 minutes. What is the ratio of the speed of the car to the speed of the bus?

Soln:- D^x

	car	:	bus
Time =	45 min	:	90 min
	1	:	2

Speed = 2 : 1 Ans.

[37] A car can finish a certain journey in 20 hours at a speed of 132 km/hr. In order to cover the same distance in 16.5 hr, the speed of the car must be increased by?

Soln

T → 200 : 16.5
40 : 33

S → 33 : 40
+7

33 $\xrightarrow{\times 4}$ 132 km/hr

7 $\xrightarrow{\times 4}$ 28 km/hr

Ans

[38] A car covers a certain distance in $9\frac{1}{3}$ hrs. If the speed is increased by 9 km/hr, then it will take 80 minutes less to cover the same distance. Find the distance covered by car?

Soln:-

$$T \rightarrow 9\frac{1}{3} \text{ hr} : 8 \text{ hr}$$

$$7 : 6$$

$$S \rightarrow 6 : 7$$

$$1 \rightarrow 9 \text{ km/hr}$$

$$7 \rightarrow 63 \text{ km/hr}$$

$$D = 63 \times 8 \text{ hr} = \boxed{504 \text{ km}} \text{ Ans}$$

[39] A man covers a certain distance in 3 hours 36 minutes if he walks at the rate of 5 km/hr. If he covers the same distance on cycle at the rate of 24 km/hr, then the time taken by him in minutes is!

Soln:-

$$S \rightarrow 5 : 24$$

$$24 \xrightarrow{\times 9} 216 \text{ min}$$

$$T \rightarrow 24 : 5$$

$$5 \xrightarrow{\times 9} \boxed{45 \text{ min}} \text{ Ans}$$

[40] A train running at $\frac{9}{13}$ of its own speed reached a place in 39 hours. How much time could be saved if the train would run at its own speed?

Soln

$$S \rightarrow 13 : 9$$

$$T \rightarrow 9 : 13 \xrightarrow{\times 3} 39 \text{ hr}$$

$$4 \rightarrow 4 \times 3 = \boxed{12 \text{ hrs}} \text{ Ans}$$

[41] A and B started at the same time from the same place for a certain destination. B walking at $\frac{5}{6}$ of A's speed reached the destination 1 hour 15 minutes after A. B reached the destination in ?

Soln:-

$$S \rightarrow \begin{array}{cc} A & : & B \\ 6 & : & 5 \end{array}$$

$$T \rightarrow \begin{array}{cc} 5 & : & 6 \\ \text{---} \ominus \text{---} & & \\ & \rightarrow & 1 \text{ hr } 15 \text{ min} \end{array}$$

$$B's \quad 6 \rightarrow 1\frac{1}{4} \times 6 = \boxed{7\frac{1}{2} \text{ hr}} \quad \underline{\text{Ans}}$$

[42] Two buses travel to a place at 45 km/hr and 60 km/hr respectively. If the second bus takes $5\frac{1}{2}$ hours less than the first for the journey, the length of the journey is:

Soln:-

$$\begin{array}{cc} A & B \\ 45 \text{ km/hr} & 60 \text{ km/hr} \end{array}$$

$$S \rightarrow 3 : 4$$

$$T \rightarrow \begin{array}{cc} 4 & : & 3 \\ \text{---} \ominus \text{---} & & \\ & \rightarrow & 5\frac{1}{2} \text{ hr} \end{array}$$

$$\text{Time Speed of A } 4 \rightarrow 22 \text{ hr}$$

$$D = 45 \times 22 = \boxed{990 \text{ km}} \quad \underline{\text{Ans}}$$

[m-I]

$$\begin{array}{ccc} \text{Time} & A & \\ 4 \leftarrow & 45 \text{ km/hr} & \\ \text{---} \ominus \text{---} & & \\ 3 \leftarrow & 60 \text{ km/hr} & B \end{array} \quad \begin{array}{l} \text{Let Distance} \\ 180 \leftarrow \text{LCM}(45, 60) \end{array}$$

$$1 \rightarrow 5\frac{1}{2} \text{ hr}$$

$$\begin{aligned} \text{Distance} &= 180 \times 5\frac{1}{2} \\ &= \boxed{990 \text{ km}} \end{aligned}$$

43 Two cars A and B travel from one city to another, at speeds of 35 km/hr and 45 km/hr respectively. If car B takes 2 hours lesser time than car A for the journey, then the distance (in kms) between the two cities is?

Solⁿ:-

	35 km/hr		45 km/hr
Speed:-	7	:	9
Time:-	9	:	7

$\underbrace{\quad\quad\quad}_{\text{2}}$ $\xrightarrow{\quad\quad\quad}$ 2 hr
 $\underbrace{\quad\quad\quad}_{\text{1}}$ $\xrightarrow{\quad\quad\quad}$ 1 hr

$$\text{Time}_A = 9 \text{ hr}$$

$$\text{Distance} = 35 \times 9 = \boxed{315 \text{ km}}$$

m-II

$$\begin{aligned}
 \text{Distance} &= \frac{v_1 \times v_2}{|v_1 - v_2|} \times t \\
 &= \frac{35 \times 45}{10} \times 2 \text{ hr} \\
 &= \boxed{315 \text{ km}} \underline{\underline{\text{Ans}}}
 \end{aligned}$$

m-III

	Time		
2 hr	9	← 35 km/hr	Let Distance 315
	7	← 45 km/hr	

$$\begin{aligned}
 2 &\xrightarrow{\quad\quad\quad} 2 \text{ hr} \\
 1 &\xrightarrow{\quad\quad\quad} 1 \text{ hr}
 \end{aligned}$$

$$\text{Distance} = \boxed{315 \text{ km}} \underline{\underline{\text{Ans}}}$$

- 44** A is twice as fast as B and B is thrice as fast as C. The journey covered by C. in $1\frac{1}{3}$ hours will be covered by A in.

Sol.:-

$$A : B : C$$

$$2 : 1 : 1$$

$$\text{Speed: } 6 : 3 : 1$$

$$\text{Time: } \frac{1}{6} : \frac{1}{3} : \frac{1}{1}$$

$$1 : 2 : 6 \longrightarrow \frac{3}{2} \text{ hr}$$

$$1 \longrightarrow \frac{1}{4} \text{ hr} = \boxed{15 \text{ min.}} \text{ Ans}$$

- 45** A person walks a distance from A to B. at 15km/hr, and from point B to A at 20km/hr. If he takes 3 hours to complete the journey, then what is the distance from point A to B ?

Sol.:-

$$S \rightarrow 15 \text{ km/hr} : 20 \text{ km/hr}$$

$$1 : 2$$

$$T \rightarrow \begin{array}{c} 2 : 1 \\ \text{---} \text{---} \text{---} \\ \text{+} \\ 3 \end{array} \longrightarrow 3 \text{ hr}$$

$$(2 \text{ hr})$$

$$\text{Distance} = 15 \text{ km/hr} \times 2 \text{ hr} = \boxed{30 \text{ km}}$$

Ans

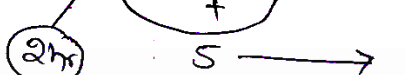
$$\text{avg Speed} = \frac{2 \times 15 \times 20}{15 + 20} = 20 \text{ km/hr}$$

$$\text{Total Distance covered} = 20 \times 3 = 60 \text{ km}$$

$$\text{Distance (A to B)} = \boxed{30 \text{ km}} \text{ Ans}$$

5] A person, who can walk down a hill at the rate of $4\frac{1}{2}$ km/hr and up the hill at the rate of 3 km/hr. He ascends and comes down to his starting point in 5 hours. How far did he ascend?

Soln:

$4\frac{1}{2} \text{ km/hr}$ 3 km/hr
 $S \rightarrow 3 : 2$
 $T \rightarrow 2 : 3$


$$\text{Distance} = 4\frac{1}{2} \text{ km/hr} \times 2 \text{ hr} = \boxed{9 \text{ km}} \text{ Ans}$$

47] A train travels 198 km in certain time. If train travels $14\frac{2}{7}\%$ faster than a car then both reach at a same time because there was a 16.5 min halt for train. Find the speed of car?

Soln:

Train Car

S → 8 : 7

T → 7 : 8

$\ominus$

 1 → 16.5 min

Time for car $\rightarrow 8 \times 16.5$
 $= 132 \text{ min}$

$$\text{Speed of car} = \frac{198 \text{ km}}{132} \times 60$$

$$= 90 \text{ km/hr} \quad \underline{\text{Ans}}$$

[48] If a person walks at 15 km/hr instead of 9 km/hr, he would have walked 3 km more in the same time. What is the actual distance (in kms) travelled by him?

Soln:-

$$\begin{array}{rcl}
 & 15 \text{ km/hr} & 9 \text{ km/hr} \\
 S \rightarrow & 5 & : 3 \\
 D \rightarrow & 5 & : 3 \\
 & \swarrow & \searrow \\
 & 2 & \rightarrow 3 \text{ km} \\
 & 1 & \rightarrow \frac{3}{2} \\
 & 5 & \rightarrow \frac{3}{2} \times 5 = \boxed{7.5 \text{ km}} \text{ Ans}
 \end{array}$$

[49] A delivery boy started from his office at 10 a.m. to deliver an article. He rode his scooter at speed of 32 km/hr. He delivered the article and waited for 15 minutes to get the payment. After the payment was made, he reached his office at 11:25 a.m., travelling at a speed of 24 km/h. Find the total distance travelled by the boy.

Soln:-

10 AM $\xrightarrow{1 \text{ hour } 25 \text{ min}}$ 11:25 AM

15 min rest

Total time = 1 hr 10 min

D^x

$$\begin{array}{rcl}
 S \rightarrow & 32 : 24 \\
 & 4 : 3 \\
 T \rightarrow & 3 : 4 \rightarrow \textcircled{40 \text{ min}} \\
 & \swarrow \quad \searrow \\
 & \textcircled{30 \text{ min}} \quad \begin{array}{l} 7 \rightarrow 1 \text{ hr } 10 \text{ min} = 70 \text{ min} \\ 1 \rightarrow 10 \text{ min} \end{array}
 \end{array}$$

$$\begin{aligned}
 D &= 32 \times \frac{30}{60} \text{ min} + 24 \times \frac{40}{60} \text{ min} \\
 &= 16 + 16 \\
 &= \boxed{32 \text{ km}} \text{ Ans}
 \end{aligned}$$

[50] If I walk at $\frac{7}{6}$ of my usual speed, then I reach my office 15 minutes early. what is the usual time taken (in minutes) by me to reach the office?

Soln:- **[m-I]**

$$S \rightarrow 6 : 7$$

$$T \rightarrow 7 : 6$$

①

$$1 \rightarrow 15 \text{ min}$$

$$\text{Usual time} = 7 \rightarrow \boxed{105 \text{ min}}$$

Ans

[m-II]

$$D = \frac{V_1 \times V_2}{V_1 - V_2} \times T$$

$$= \frac{6 \times 7}{1} \times \frac{15}{60} \leftarrow \text{for hr}$$

$$D = \frac{21}{2} \text{ km}$$

$$T = \frac{\frac{21}{2}}{\frac{2}{6}} \times 60 = \boxed{105 \text{ km}}$$

[51] Walking at 60% of his usual speed, a man reaches his destination 1 hour 40 minutes late. His usual time (in hours) to reach the destination is:

Soln $60\% = \frac{3}{5}$

$$S \rightarrow 5 : 3$$

$$T \rightarrow 3 : 5$$

①

$$2 \rightarrow 1 \frac{40}{60} = \frac{5}{3}$$

$$1 \rightarrow \frac{5}{6}$$

Usual time

$$3 \times \frac{5}{6} = \boxed{2 \frac{1}{2} \text{ hr}} \quad \underline{\underline{\text{Ans}}}$$

[52] With an average speed of 40 km/hr a train reaches its destination on time. If it goes with an average speed of 35 km/hr, it is late by 15 minutes. The total journey is?

Soln:-

$$D = \frac{40 \times 35}{5} \times \frac{15}{60}$$
$$= \boxed{70 \text{ km}} \underline{\underline{\text{Ans}}}$$

[53] A person covers a certain distance with a certain speed if he reduce his speed by 25 km/hr. Then he will 12.5 min late. If his initial speed is 160 km/hr. Find the distance covered by him.

Soln:-

$$D = \frac{160 \times 135}{25} \times \frac{12.5}{60}$$
$$= \boxed{180 \text{ km}} \underline{\underline{\text{Ans}}}$$

[54] Anita travels from house at $3\frac{1}{2}$ km/hr and reaches her school 6 minutes late. The next day she travels at $4\frac{1}{2}$ km/h and reaches her school 10 minutes early. What is the distance her house the school,

[SSCCGL Mams
2020]

Soln

$$D = \frac{\frac{7}{2} \times \frac{9}{2}}{1} \times \frac{(6+10)}{60}$$
$$= \frac{63}{4} \times \frac{16}{60}$$
$$= \boxed{4.2 \text{ km}} \underline{\underline{\text{Ans}}}$$

55 A person covers a distance of 300 km and then returns to the starting point. The time taken by him for the outward journey is 5 hours more than the time taken for the return journey. If he returned at a speed of 10 km/h more than the speed of going. What was the speed (in km/h) for the outward journey? [SSC CPO-2020]

Soln:-

$$D = \frac{v_1 v_2}{|v_1 - v_2|} \times T$$

$$300 = \frac{v(v+10)}{10} \times 5$$

$$600 = v(v+10)$$

$$v = \boxed{20 \text{ km/hr}} \text{ Ans}$$

56 A motocyclist left $6\frac{6}{9}$ minutes later than the scheduled time but in order to reach his destination 21 km away in time, he had to increase his speed by 12 km/hr from the usual speed. What is usual speed (in km/hr) of the motocyclist?

Soln

$$21 = \frac{v(v+12)}{12} \times \frac{60}{9 \times 60}$$

$$21 \times 12 \times 9 = v(v+12)$$

$$42 \times 54 = v(v+12)$$

$$v = \boxed{42 \text{ km/hr}} \text{ Ans}$$

57 A truck covers a distance of 384 km at a certain speed. If the speed is decreased by 16 km/h. It will take 2 hours more to cover the same distance. the 75% of its original speed (in km/hr) is:

Soln:-

$$384 = \frac{v(v-16)}{16} \times 2$$

$$64 \times 48 = v(v-16)$$

$$v = 64 \text{ km/hr}$$

$$75\% = 64 \times \frac{3}{4} = \boxed{48 \text{ km/hr}} \text{ Ans}$$

58 A covered a distance of 240 km at a certain speed. Had his speed been 8 km/hr less, then the time taken would have been one hour more for covering the same distance. How much time (in hours) will he takes to cover a distance of 480 km at his original speed?

Soln:-

$$240 = \frac{v(v-8)}{8} \times 1$$

$$240 \times 8 = v(v-8)$$

$$48 \times 40 = v(v-8)$$

$$v = 48 \text{ km/hr}$$

$$\text{Time} = \frac{480}{48} = \boxed{10 \text{ hrs}} \text{ Ans}$$

59 Due to inclement weather, an airplane reduced its speed by 300km/hr and reached the destination of 1200km late by 2hrs. Then the schedule duration of the flight was.

Soln:-

$$1200 = \frac{V(V-300)}{300} \times 2$$

$$600 \times 300 = V(V-300)$$

$$V = 600 \text{ km/hr}$$

$$t = \frac{1200}{600} = \boxed{2 \text{ hour}} \quad \underline{\underline{\text{Ans}}}$$

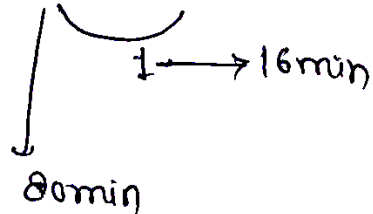
60 A person covers a certain distance with certain speed if he increases his speed by 15km/hr. Then he will be 16min. early. By how much time he will be late if he reduce his speed by 12km/hr, if his initial speed is 60km/hr.

Soln

$$S \rightarrow 60 : 75$$

$$4 : 5$$

$$T \rightarrow 5 : 4$$

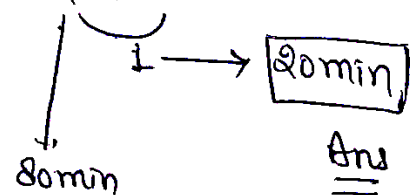


after

$$S \rightarrow 60 : 48$$

$$5 : 4$$

$$T \rightarrow 4 : 5$$



Ans

Q1 If a person reduces his speed by 27.27%, he takes 66 min. to reach his destination. Find the time taken by the person (in min.) to reach his destination if he increases his speed by 20%.

Soln:- $27.27\% = \frac{3}{11}$

$20\% = \frac{1}{5}$

$S \rightarrow 11 : 8$

$S \rightarrow 5 : 6$

$T \rightarrow 8 : 11$

$T \rightarrow 6 : 5$

$\downarrow \quad \downarrow$
48 66 min

$\downarrow \quad \downarrow$
48 40 min Ans

Q2 A man cycles at the speed of 8 km/hr and reaches office at 11 am and when he cycles at the speed of 12 km/hr, he reaches office at 9 am. At what speed should he cycle so that he reaches his office at 10 am?

Soln:-

Speed = 8 km/hr 12 km/hr

$S \rightarrow 2 : 3$

$T \rightarrow 3 : 2$

$\swarrow \quad \searrow$
6 hr 4 hr
1 $\rightarrow$ 2 hr

11 am - 9 am
2 hr

Starting time :- 11 am - 6 hr
= 5 am

9 am - 4 hr
5 am

Distance = 8 km/hr $\times$ 6 hr = 48 km

Speed = $\frac{48}{5} = \boxed{9.6 \text{ km/hr}}$ Ans

5 am - 10 am
= 5 hr

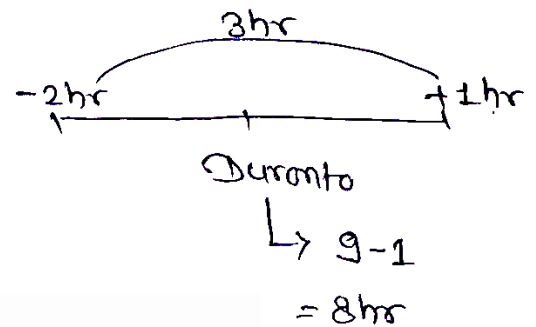
[63] To travel 432 km, an Express train takes 1 hour more than Duronto. If however, the speed of Express train is increased by 50%, it takes 2 hours less than Duronto. What is the speed (in km/hr) of Duronto train?

Soln:-

$$S_E : 2 : 3$$

$$T_E \rightarrow 3 : 2$$

$\swarrow$ $\searrow$
 9hr $1 \rightarrow 3\text{hr}$



$$\text{The Speed of Duronto} = \frac{432}{8}$$

$$= \boxed{54 \text{ km/hr}} \text{ Ans}$$

[64] To cover a distance of 416 km, a train A takes $2\frac{2}{3}\text{hr}$ more than train B. If the speed of A is doubled, it would take $1\frac{1}{3}\text{hr}$ less than B. What is the speed (in km/hr) of train A?

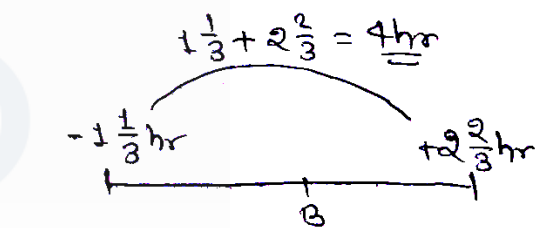
Soln:-

A before after

Speed :- 1 : 2

Time: 2 : 1

$\swarrow$ $\searrow$
 8hr $1 \rightarrow 4\text{hr}$



$$\text{The speed of A} = \frac{416 \text{ km}}{8\text{hr}}$$

$$= \boxed{52 \text{ km/hr}} \text{ Ans}$$

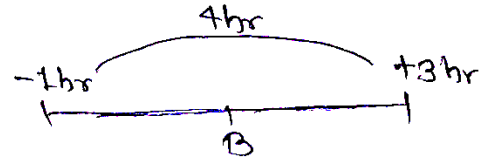
65 A takes 3 hours more than B to walk 'd' km. If A doubles his speed, then he can make it in 1 hour less than B. How much time (in hours) does A require to walk 'd' km?

Soln:-

Speed ^A :- 1 : 2

Time :- 2 : 1
 $\swarrow$ $\times 4$ $\searrow$
 1 $\rightarrow$ 4hr

8hr Ans



66 A takes 2 hours 30 minutes more than B to walk 40km. If A doubles his speed, then he can make it in 1 hour less than B. What is the average time taken by A and B to walk a 40km distance? [SSC CPO - 2020]

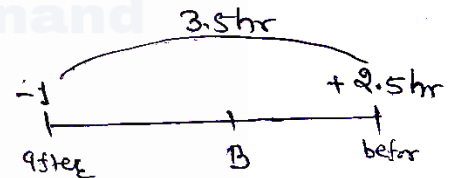
Soln:-

Speed ^A $\rightarrow$ 1 : 2

Time $\rightarrow$ 2 : 1
 $\swarrow$ $\searrow$
 1 $\rightarrow$ 3.5hr

A $\rightarrow$ 7hr

B $\rightarrow$ 7hr - 2.5hr
 $=$ 4.5hr



$$\text{avg time} = \frac{7 + 4.5}{2} = \frac{11.5}{2} \text{ hr}$$

$=$ **5 hour 45 minute** Ans

- Q7** A takes 2 hours more than B to cover a distance of 40 km. If A doubles his speed, he takes $1\frac{1}{2}$ hours more than B to cover 80 km. To cover a distance of 90 km, How much time will B take travelling at his same speed?

[SSC CGL Mains 2020]

Soln:-

A $\rightarrow$ 40km distance $\rightarrow$ +2hr more
80km distance $\rightarrow$ +4hr more

A
Speed $\rightarrow 1 : 2$

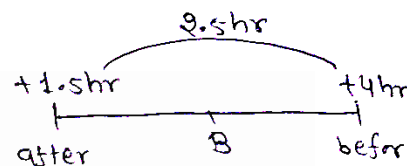
Time $\rightarrow 2 : 1$

$\swarrow$ $\searrow$
Time A = 5hr

Time B = 5 - 4 = 1hr

Speed of B = $\frac{80\text{km}}{1\text{hr}} = 80\text{km/hr}$

Time = $\frac{90\text{km}}{80\text{km/hr}} = \frac{9}{8}\text{hr} = \boxed{1\frac{1}{8}\text{hr}}$ Ans



- Q8** Train A takes 45 minutes more than train B to travel a distance of 450 km. Due to engine trouble speed of train B falls by a quarter, so it takes 30 minutes more than Train A to complete same journey. What is the speed of Train A (in km/hr)?

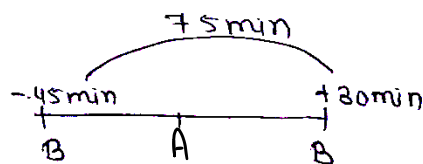
Soln:-

Speed B $\rightarrow 4 : 3$

Time B $\rightarrow 3 : 4$

$\swarrow$ $\searrow$
 3×75
= 225 min

Speed A = $\frac{450}{4.5} = \boxed{100\text{km/hr}}$ Ans



$\rightarrow 225 + 45 = 270\text{min}$
= 4.5hr

69 Flight A usually takes 1 hour more than flight B to travel a distance of 7200 km. Due to engine trouble speed of flight B falls by a factor of $\frac{5}{6}$ th, so it takes 36 minutes more than flight A to complete the same journey? What is the speed of flight A (in km/hr)?

Soln:-

$$\text{Speed}_B :- 6 : 5$$

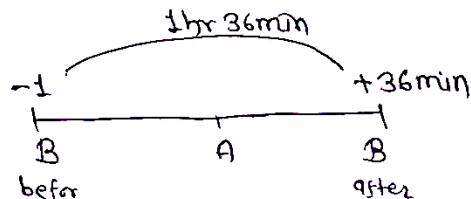
$$\text{Time}_B \rightarrow 5 : 6$$

$$\downarrow \quad \quad \quad \downarrow \rightarrow 1 \text{ hr } 36 \text{ min} = \frac{8}{5} \text{ hr}$$

$$\text{Time}_B \rightarrow 8 \text{ hr}$$

$$\text{Time}_A \rightarrow 8 + 1 = 9 \text{ hr}$$

$$\text{Speed}_A = \frac{7200 \text{ km}}{9 \text{ hr}} = \boxed{800 \text{ km/hr}} \quad \text{Ans}$$



70 A bus travels $\frac{2}{5}$ of a total journey at its usual speed. The remaining distance was covered by bus at $\frac{6}{7}$ of its usual speed. Due to slow speed it reaches its destination 50 minutes late. If the total distance is 200 km then what is the usual speed (in km/hr) of bus?

Soln:-

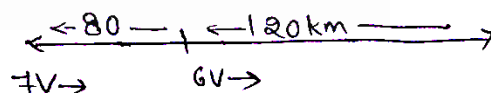
$$\frac{20 \cdot 120}{6V} - \frac{80}{7V} = \frac{50}{60}$$

$$\frac{140 - 80}{7V} = \frac{5}{6}$$

$$\frac{60}{7V} = \frac{5}{6}$$

$$7V = \frac{360}{5}$$

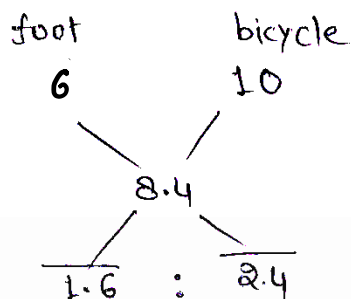
$$\text{usual speed} = 7V = \boxed{72 \text{ km/hr}} \quad \text{Ans}$$



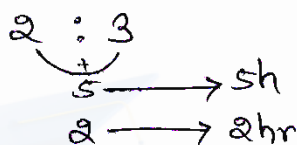
71 A man travelled distance of 42km in 5 hours. He travelled Partly on foot at the rate of 6km/hr and the rest on bicycle at the rate of 10km/hr. The distance travelled on foot is;

[SSC CGL Mains-2020]

Soln:- Alligation method:-



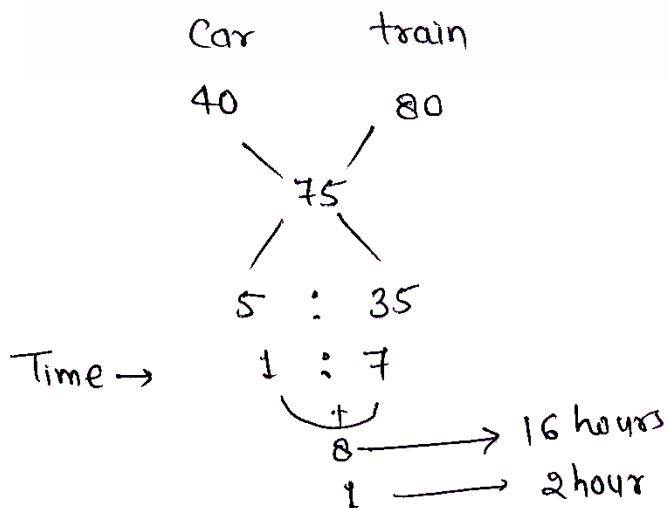
ratio of Time:-



$$\text{distance} = 6\text{km/hr} \times 2 = \boxed{12\text{km}} \text{ Ans}$$

72 A man travelled a distance of 1200 km in 16 hours, He travelled partly by car at a speed of 40km/hr, and partly by train at a speed of 80km/h what is the distance traveled by car?

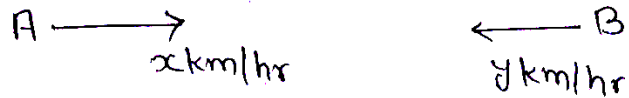
Soln:-



$$\begin{aligned} \text{Distance} &= 40\text{km/hr} \times 2\text{ hour} \\ &= \boxed{80\text{km}} \text{ Ans} \end{aligned}$$

Relative Speed

(I)



$$\text{Relative speed} = (x + y) \text{ km/hr}$$

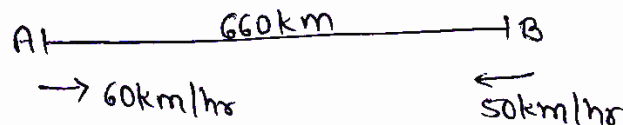
(II)



$$\text{Relative speed} = (x - y) \text{ km/hr}$$

[73] Distance between A & B is 660 km. Two person from A and B with speed 60 km/hr and 50 km/hr start to meet. Find meeting time.

Solⁿ



$$\text{Time} = \frac{\text{Distance}}{\text{Relative speed}}$$

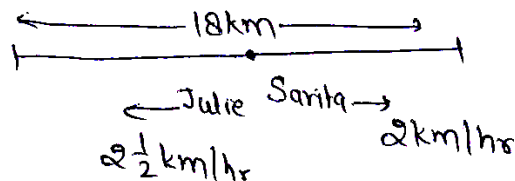
$$= \frac{660}{(60 + 50)}$$

$$= \frac{660}{110}$$

$$= \boxed{6 \text{ hr}} \quad \underline{\underline{\text{Ans}}}$$

74 Sarita and Julie start walking from the same place in the opposite directions. If Julie walks at a speed of $2\frac{1}{2}$ km/h and Sarita at a speed of 2 km/hr, in how much time will they be 18 km apart?

Soln:-



$$\text{Time} = \frac{18}{(2\frac{1}{2} + 2)} = \frac{18}{\frac{9}{2}} = \boxed{4\text{hr}} \quad \underline{\underline{\text{Ans}}}$$

75 Two cyclists P and Q cycle at 20 km/hr and 16 km/hr towards each other respectively, what was the distance (in kms) between them when they started If they meet after 40 minutes?

Soln



$$\text{Time} = \frac{\text{Distance}}{\text{Relative Speed}}$$

$$\frac{40}{60} = \frac{D}{36}$$

$$\boxed{D = 24 \text{ km}} \quad \underline{\underline{\text{Ans}}}$$

76 A and B are 20km apart. A can walk at an average speed of 4km/hr and B at 6km/hr. If they start walking towards each other at 7am, when will they meet?

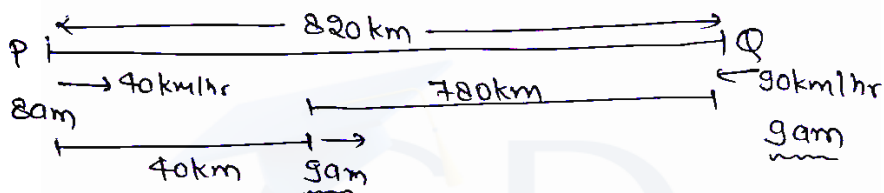
Soln:-

$$\text{Time} = \frac{20}{(4+6)} = 2\text{hr}$$

$$7\text{am} + 2\text{hr} = \boxed{9\text{am}} \text{ Any}$$

77 Distance between A and B is 820 km. Two person A and B having speed 40km/hr and 90km/hr. start to meet each others P start at 8 o'clock and Q start 9 o'clock. At what time will they meet.

Soln



$$t = \frac{780}{130} = 6\text{hr}$$

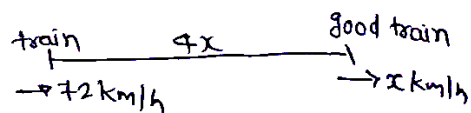
$$9\text{am} + 6 = \boxed{3\text{ बजे (am)}} \text{ Any}$$

78 4 hrs after a goods train passed a station, another train travelling at a speed of 72 km/hr following that goods train. Passed through that station. If after passing the station the train overtakes the goods train in 8 hours. What is the speed of the goods train?

Soln:-

m-I let the velocity (speed) of good train is x
in 4 hours covered distance = $4x$

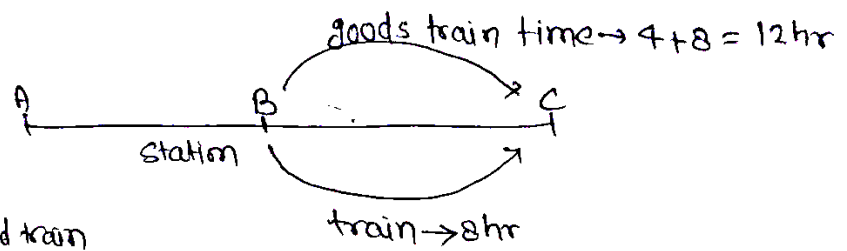
$$\text{Time} = \frac{\text{Distance}}{\text{Relative speed}}$$



$$8 = \frac{4x}{72-x}$$

$$\boxed{x = 48\text{ km/hr}} \text{ Any}$$

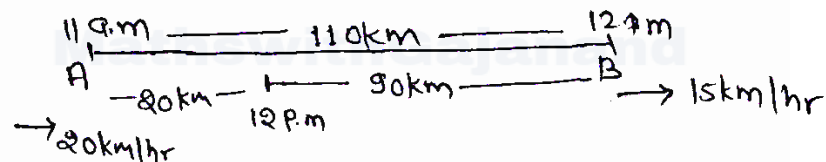
m-II Ratio method



	Train	Good train	
Time $\rightarrow$	8	: 12	
	2	: 3	
Speed $\rightarrow$	3	: 2	
	$\downarrow \times 24$	$\swarrow \times 24$	
Given	72 km/hr		
			48 km/hr <u>Ans</u>

[79] Two people A and B are at a distance of 110 km from each other at 10:30 AM. After 30 mins, A starts moving towards B at a speed of 20 km/hr while at 12 PM, B starts moving away from A at a speed of 15 km/hr. At what time (in AM) will they meet on the next day?

Soln:-



$$\text{In 1 hour} \Rightarrow 20 \times 1 = 20 \text{ km}$$

$$\text{Remaining distance} = 110 - 20 = 90 \text{ km}$$

$$\text{meeting time} = \frac{90}{(20 - 15)}$$

$$= 18 \text{ hr}$$

$$12 \text{ PM} + 18 \text{ hr} = \boxed{6 \text{ AM}} \text{ Ans}$$

[80] The driver of a car, which is travelling at a speed of 75 km/hr locates a bus 80 m ahead of him, travelling in the same direction. After 15 seconds, he finds the bus is 40 m behind the car. What is the speed of the bus (in km/hr)

[CGL Mains-2020]

Soln:-

$$\text{Time} = \frac{\text{distance}}{\text{Relative speed}}$$

$$\frac{15}{60 \times 60} = \frac{(80 + 40)}{1000(75 - x)}$$

$$(75 - x) = \frac{120 \times 4 \times 60}{1000}$$

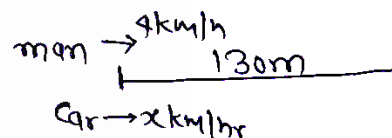
$$75 - x = 28.8$$

$$x = \boxed{46.2 \text{ km/hr}} \text{ Ans}$$

[81] A car moving in the morning fog passes a man walking at 4 km/h in the same direction. The man can see the car for 3 minutes and visibility is upto a distance of 130 m . The speed of the car is:

Soln:-

$$\text{Time} = \frac{\text{distance}}{\text{Relative speed}}$$



$$\frac{3}{60} = \frac{130}{1000 \times (x - 4)}$$

$$(x - 4) = \frac{13 \times 2}{10}$$

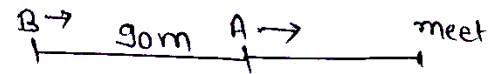
$$(x - 4) = 2.6$$

$$x = \boxed{6.6 \text{ km/hr}} \text{ Ans}$$

82 A starts from a point at a speed of 30 m/sec. After 3 seconds, B starts chasing A from the same point with a speed of 50 m/sec, what will be the total distance (in m) travelled by A and B before A is caught by B?

Soln:- **m-I**

A covers distance in 3 seconds
 $= 30 \times 3 = 90 \text{ m}$



meeting time $= \frac{90}{(50-30)} = \frac{9}{2}$ Seconds

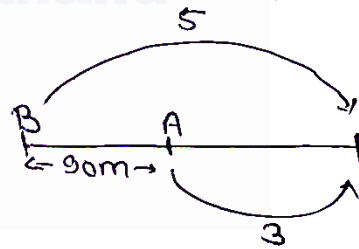
A will cover $= 30 \times \frac{9}{2} = 135 \text{ m}$

B will cover $= 50 \times \frac{9}{2} = 225 \text{ m}$

Total covered distance $= 90 + 135 + 225 = \boxed{450 \text{ m}}$ Ans

m-II

$\begin{array}{l} \text{Tx} \quad S \rightarrow \quad \begin{array}{cc} A & B \\ 30 & : & 50 \\ 3 & : & 5 \end{array} \\ \quad \quad \quad \text{D} \rightarrow \quad \begin{array}{c} 3 : 5 \\ \text{---} \ominus \text{---} \\ 2 \rightarrow 90 \text{ m} \\ 1 \rightarrow 45 \text{ m} \end{array} \end{array}$



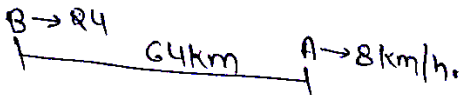
$B = 45 \times 5 = 225$

$A = 45 \times 3 = 135$

Total Distance $= 90 + 135 + 225$
 $= \boxed{450 \text{ m}}$ Ans

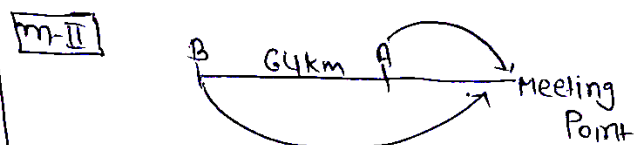
Q3 A walks at a uniform speed of 8 km/hr and 8 hours after his start, B starts on his cycle after him at speed of 24 km/hr. How far (in km) from the starting point will B catch A?

Soln: **m-I**

$$A = 8 \text{ km/hr} \times 8 \text{ hr} = 64 \text{ km}$$


$$\text{meeting time} = \frac{64}{24-8} = 4 \text{ hr}$$

$$\text{Distance} = 24 \text{ km/hr} \times 4 \text{ hr} = \boxed{96 \text{ km}} \text{ Ans}$$



$$S \rightarrow \begin{matrix} B & A \\ 24 & : & 8 \\ 3 & : & 1 \end{matrix}$$

$$D \rightarrow \begin{matrix} 3 & : & 1 \\ \text{②} & & \\ 2 & \rightarrow & 64 \text{ km} \\ 1 & \rightarrow & 32 \text{ km} \end{matrix}$$

$$\text{from B} \quad 3 \rightarrow 32 \times 3$$

$$\boxed{96 \text{ km}} \text{ Ans}$$

Q4 A, B, C walk 1 km in 5 minute, 8 minute and 10 minute respectively, C starts walking from a point, at a certain time, B starts from the same point 1 minutes later and A starts from the same point 2 minutes later then C, then A meets B and C after.

Soln Speed of A, B and C = $\frac{1000}{5}$, $\frac{1000}{8}$, $\frac{1000}{10}$

$$= 200 \text{ m/min}, 125 \text{ m/min}, 100 \text{ m/min}$$

Distance travelled by B and C before A starts

$$= 125 \text{ m}, 200 \text{ m}$$

Time taken by A to meet B and C

$$= \frac{125}{200-125}, \frac{200}{200-100}$$

$$= \frac{5}{3} \text{ min}, 2 \text{ min} \text{ Ans}$$

85 Points 'A' and 'B' are 70km apart on a highway and two cars start at the same time. If they travel in the same direction, they meet in 7 hours, but if they travel towards each other they meet in one hour. Find the speed of the two cars (in km/hr)

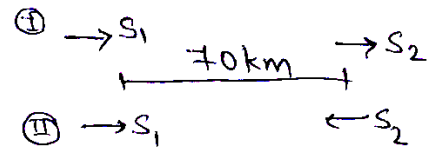
Soln:- (i) $\frac{70}{S_1 - S_2} = 7$

$$S_1 - S_2 = 10 \quad \text{--- (1)}$$

(ii) $\frac{70}{S_1 + S_2} = 1$

$$S_1 + S_2 = 70 \quad \text{--- (2)}$$

from (1) & (2) $S_1 = \boxed{40 \text{ km/hr}}$, $S_2 = \boxed{30 \text{ km/hr}}$ Ans



86 A and B are 15km apart and when travelling towards each other meet after half an hour where as they meet two and a half hour later if they travel in the same direction. The faster of the two travels at the speed of

Soln:-

$$\frac{15}{S_1 + S_2} = \frac{1}{2}$$

$$S_1 + S_2 = 30 \quad \text{--- (1)}$$

$$\frac{15}{S_1 - S_2} = \frac{5}{2}$$

$$S_1 - S_2 = 6 \quad \text{--- (2)}$$

$$S_1 = \boxed{18 \text{ km/hr}}$$

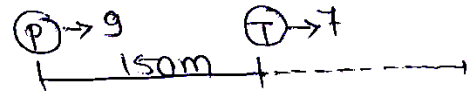
Ans

$$S_2 = \boxed{12 \text{ km/hr}}$$

[87] A thief stopped by a policeman from a distance of 150 metres. When the policeman starts the chase, the thief also starts running. Assuming the speed of the thief as 7 km/hr and that of policeman as 9 km/hr, how far the thief would have run, before he is over-taken by the policeman?

Soln:-

Time of meeting



$$= \frac{150 \times 18}{(9-7) \times 5} = 15 \times 18 \text{ sec}$$

$$\text{Distance} = 7 \times \frac{5}{18} \times 15 \times 18$$

$$= \boxed{525 \text{ m}} \text{ Ans}$$

[88] A gun is fired from a fort. A man hears the sound 10 seconds later. If the sound travels at the rate of 330 m/sec, find the distance between the fort and the man.

Soln:-

$$D = 330 \times 10$$

$$= 3300 \text{ m}$$

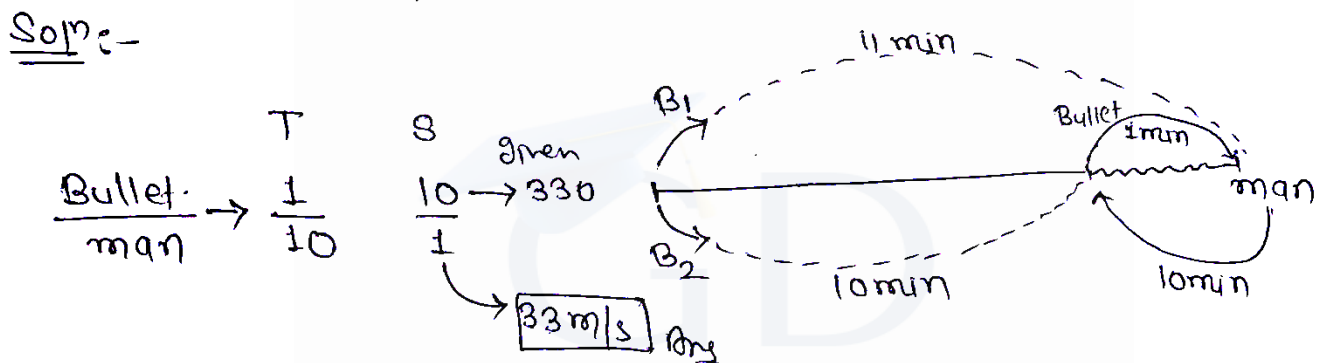
$$= \boxed{3.3 \text{ km}} \text{ Ans}$$

89 A gun is fixed at a distance of 1.7 km from Ram and he hears the sound after 25 seconds. The speed of sound in meter per second is?

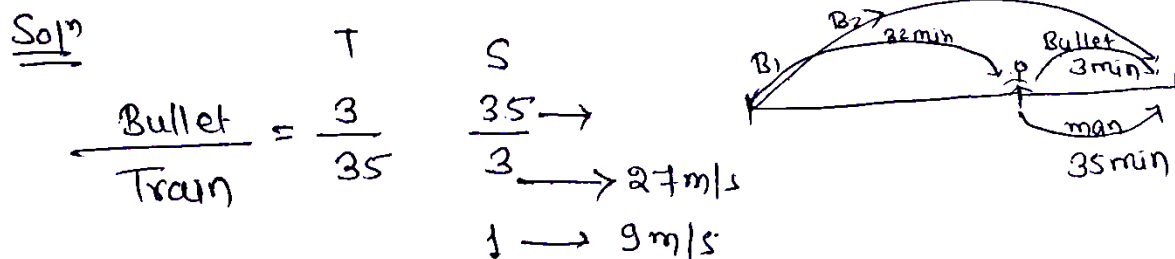
Soln:-

$$\text{Speed} = \frac{1700\text{m}}{25\text{sec}} = \boxed{68\text{m/sec}} \text{ Any}$$

90 Two bullets are fired from a place at an interval of 11 min. A person approaching that hear it at the gap of 10 min. If Speed of sound is 330 m/sec then what is the speed of person.



91 Two guns were fired from same place at an interval of 32 minutes. But a man sitting in the train going away from the place hears the 2nd firing 35 minutes after the 1st. If the speed of train is 97.2 km/hr. Find the speed of sound (in m/s)?

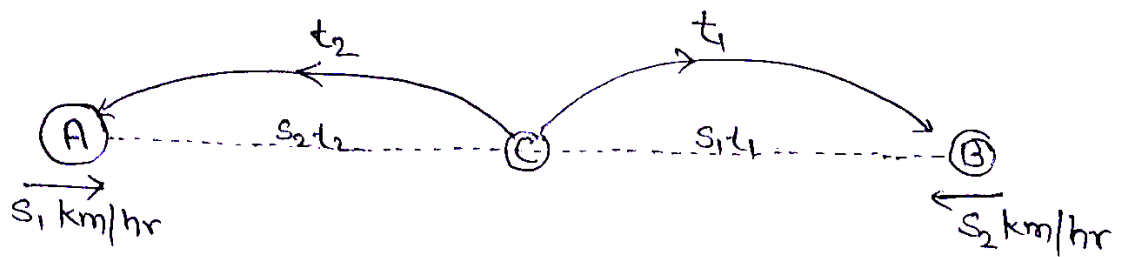


Sound 35 $\rightarrow 35 \times 9$

$$\Rightarrow \boxed{315\text{m/sec}} \text{ Any}$$

$$97.2 \times \frac{5}{18} = 27\text{m/s}$$

Concept:-



After meeting each other they take t_1 & t_2 time to reach their destination respectively.

Time^x D & S

$$\Rightarrow \frac{S_2 t_2}{S_1 t_1} = \frac{S_1}{S_2} \Rightarrow \frac{S_1^2}{S_2^2} = \frac{t_2}{t_1}$$

$$\boxed{\frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}}$$

$\Rightarrow$ If they meet after ' t ' hours

$$t = \frac{S_2 t_2}{S_1} = \sqrt{\frac{t_2}{t_1}} \times t_2$$

$$\boxed{t = \sqrt{t_1 t_2}} \leftarrow \text{meeting time}$$

$\Rightarrow$ Distance between A to B

$$\boxed{D = S_1 t_1 + S_2 t_2}$$

92 Two trains started at the same time, one from A to B and the other from B to A. If they arrived at B and A respectively 4 hours and 9 hours after they passed each other, the ratio of the speeds of the two trains was:

Soln:-

$$\frac{S_A}{S_B} = \sqrt{\frac{t_2}{t_1}}$$

$$\frac{S_A}{S_B} = \sqrt{\frac{9}{4}} = \frac{3}{2} \text{ Ans}$$

93 A and B start moving from places X to Y and Y to X, respectively, at the same time on the same day. After crossing each other, A and B take $5\frac{4}{9}$ hours and 9 hours, respectively, to reach their destinations. If the speed of A is 33 km/h, then the speed (in km/h) of B is:

Soln:-

$$\frac{33}{S_B} = \sqrt{\frac{9 \times 9}{49}}$$

$$\frac{33}{S_B} = \frac{9}{7} \Rightarrow S_B = \boxed{25\frac{2}{3} \text{ km/hr}} \text{ Ans}$$

94 A and B are travelling towards each other from the points P and Q respectively. After crossing each other, A and B take $6\frac{1}{8}$ hours and 8 hours respectively. If the speed of B is 16.8 km/h, then the speed (in km/h) of A is?

Soln:-

$$\frac{S_1}{16.8} = \sqrt{\frac{8 \times 8}{49}}$$

$$\frac{S_1}{16.8} = \frac{8}{7}$$

$$S_1 = \boxed{19.2 \text{ km/hr}} \text{ Ans}$$

95 A and B start moving towards each other from places X and Y, respectively, at the same time on the day. The speed of A is 20% more than of B. After meeting on the way, A and B take p hours and $7\frac{1}{5}$ hours, respectively. What is the value of p ?

Soln:-

$$\begin{array}{ccc} & A & B \\ \text{Speed} \rightarrow & 6 & : & 5 \end{array}$$

$$\frac{6}{5} = \sqrt{\frac{36}{5p}}$$

$$\frac{36}{25} = \frac{36}{5p}$$

$$\boxed{p = 5} \text{ Ans}$$

96 Aman and Kapil start from Delhi and Gwalior respectively towards each other at same time. They meet at Mathura and then take 196 minutes and 225 minutes respectively to reach Gwalior and Delhi. If speed of Aman is 30 km/hr, the speed (km/hr) of Kapil?

Soln:-

$$\frac{s_1}{s_2} = \sqrt{\frac{t_2}{t_1}}$$

$$\frac{30}{s_2} = \sqrt{\frac{225}{196}}$$

$$\frac{30}{s_2} = \frac{15}{14}$$

$$s_2 = \boxed{28 \text{ km/hr}} \text{ Ans}$$

Advanced Question

- 97** A train leaves a station A at 7 am and reaches another station B at 11 am. Another train leaves B at 8 am and reaches A at 11:30 am. The two trains cross on another at ?

Soln:-

D^x

Time $\rightarrow 4 : 3\frac{1}{2}$

$\rightarrow 4 : \frac{7}{2}$

Speed $\rightarrow 07 : 8$

$7x : 8x$

Total distance = $7x \times 4\text{hr} = 28x$

Distance covered in 1 hr

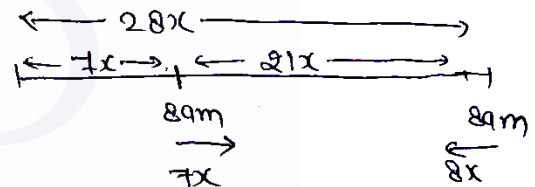
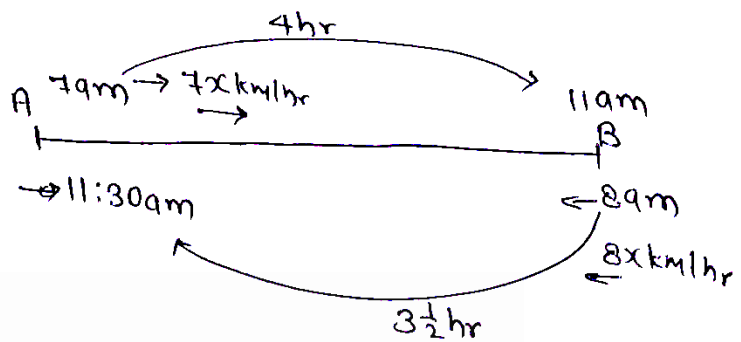
$7x \times 1\text{hr} = 7x$

meeting time = $\frac{21x}{(7x+8x)}$

$= \frac{21}{15} = 1\text{hr } 24\text{min}$

meeting time = 8 am + 1hr 24min

$= \boxed{9:24\text{ am}} \text{ Ans}$



- 98** A train leaves Delhi at 10 am and reaches Jaipur at 4 p.m. on same day. Another train leaves Jaipur at 12 p.m. and reaches Delhi at 5 p.m. on same day. What is the time of day (approximately) when the two trains will meet?

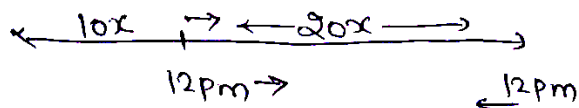
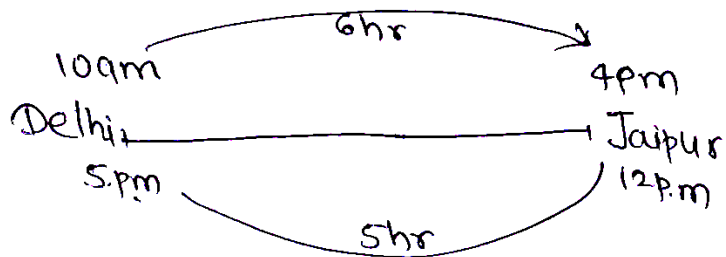
Time $\rightarrow 6 : 5$

Speed $\rightarrow 5 : 6$

Let $5x : 6x$

$$\begin{aligned}\text{Total Distance} &= 5x \times 6\text{hr} \\ &= 30x\end{aligned}$$

In 2 hr 1st train covered
distance = $10x$

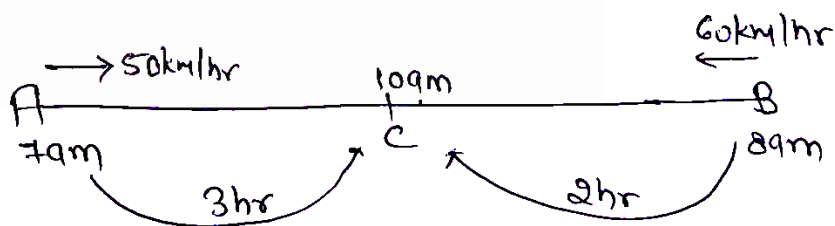


$$\text{meeting time} = \frac{20x}{(5x+6x)} = \frac{20}{11} = 1\text{hr } 49\text{min}$$

$$= 12\text{pm} + 1\text{hr } 49\text{min} = \boxed{1:49\text{ PM}} \text{ Ans}$$

99 A train starts from A at 7 a.m. towards B with speed 50 km/hr. Another train starts from B at 8 a.m. with speed 60 km/hr towards A. Both of them meet at 10 a.m. at C. The ratio of the distance AC to BC is?

Soln.:-



$$AC = 50\text{km/hr} \times 3\text{hr} = 150$$

$$BC = 60\text{km/hr} \times 2\text{hr} = 120$$

$$AC : BC = 150 : 120$$

$$= \boxed{5 : 4} \text{ Ans}$$

100 The distance between two stations A and B is 200 km. A train runs from A to B at a speed of 75 km/hr, while another train runs from B to A at a speed of 85 km/hr. What will be the distance between the two train (in km) 3 minutes before they meet? [CGL Pre-2022]

Soln:-

$$\begin{aligned} \text{Distance} &= \text{Relative speed} \times \text{Time} \\ &= (75 + 85) \times \frac{3}{60} \\ &= 160 \times \frac{3}{60} \\ &= \boxed{8 \text{ km}} \text{ Ans} \end{aligned}$$

101 A train can travel 25% faster than a car. Both start from Point A at the same time and reach point B 165 km away at the same time. On the way the train takes 40 min. for stopping at the stations. What is the difference in the speeds (in km/hr) of train and car?

Soln:-

	Car	train	
Speed $\rightarrow$	4	5	
Time $\rightarrow$	5	4	$\times 40$
		1	$\rightarrow 40 \text{ min}$
	200 min.		160 min

$$25\% = \frac{1}{4}$$

$$\text{Speed of Car} = \frac{165 \times 60}{200} = 49.5 \text{ km/hr}$$

$$\text{Speed of train} = \frac{165 \times 60}{160} = 61.875 \text{ km/hr}$$

$$\begin{aligned} \text{Difference} &= (61.875 - 49.5) \\ &= \boxed{12.375 \text{ km/hr}} \text{ Ans} \end{aligned}$$

102 B starts 4 minutes after A from the same point, for a place at a distance of 7 miles from the starting Point. A on reaching the destination turns back and walks a mile where he meets B. If A's speed is a mile in 8 minutes then B's speed is a mile in ---- minutes?

Soln:-

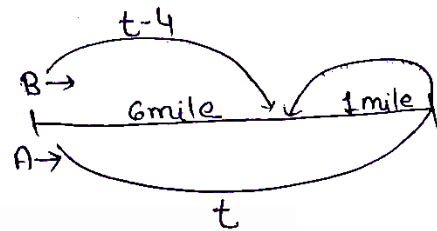
$$\begin{aligned}\text{Total Distance travelled by A} \\ &= 7 \text{ mile} + 1 \text{ mile} \\ &= 8 \text{ mile}\end{aligned}$$

$$\begin{aligned}\text{Time} &= 8 \text{ mile} \times 8 \text{ min} \\ t &= 64 \text{ min.}\end{aligned}$$

$$\text{Time taken by B} = t - 4 = 64 - 4 = 60 \text{ min.}$$

$$\begin{aligned}\text{Speed} &= \frac{6 \text{ mile}}{1 \text{ mile}} \xrightarrow{\text{in}} \frac{60 \text{ min}}{6} = 10 \text{ min}\end{aligned}$$

$$\text{Speed} = \boxed{10 \text{ min/mile}} \text{ Ans}$$



103 From two places, 60 km apart A and B start towards each other at the same time and meet each other after 6 hours. If A travelled with $\frac{2}{3}$ of his usual speed and B travelled with double of his speed they would have met after 5 hours. The speed of A is:

Soln:-

$$\text{Relative speed} = \frac{\text{Distance}}{\text{Time}}$$

$$S_1 + S_2 = \frac{60}{6} = 10 \quad \text{--- (1)}$$

$$\frac{2}{3}S_1 + 2S_2 = \frac{60}{5} = 12 \quad \text{--- (2)}$$

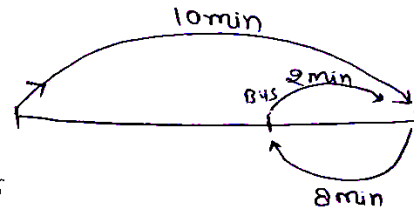
from ① & ②

$$S_1 = \boxed{6 \text{ km/hr}} \text{ Ans}$$

- 104** Buses start from a bus terminal with a speed of 20 km/hr at intervals of 10 minutes. What is the speed of a man coming from the opposite direction towards the bus terminal if he meets the buses at intervals of 8 minutes?

Soln:-

	Time	Speed	
Buses	10	20	Given
man	8	$\frac{8}{2} = \frac{4}{1}$	$\times 5 \rightarrow 20 \text{ km/hr}$
			$\times 5 \rightarrow \boxed{5 \text{ km/hr}}$ <u>Ans</u>



- 105** A car travels from P to Q at a constant speed. If its speed were increased by 10 km/hr, it would have taken one hour lesser to cover the distance. It would have taken further 45 min lesser if the speed was further increased by 10 km/hr. The distance between the two cities is?

Soln:- **m-f**

$$① = \frac{V_1 V_2}{(V_1 - V_2)} \times T$$

$$① = \frac{v(v+10)}{10} \times 1 = \frac{v(v+20)}{20} \times \frac{7}{4}$$

$$v = 60 \text{ km/hr}$$

$$① = \frac{60 \times 70}{10} \times 1 = \boxed{420 \text{ km}}$$
 Ans

m-11

$$\begin{array}{lcl} S & \times & T \\ +10 & -1 & \rightarrow 10 \\ +20 & -\frac{7}{4} & \rightarrow 35 \end{array} \quad d = s \times t$$

$$-S + 10t = 10 \quad \text{--- (1)}$$

$$-\frac{7S}{4} + 20t = 35 \quad \text{--- (2)}$$

Solve

$$S = 60 \text{ km/hr}$$

$$t = 7 \text{ hour}$$

$$① = 60 \times 7 = \boxed{420 \text{ km}}$$
 Ans

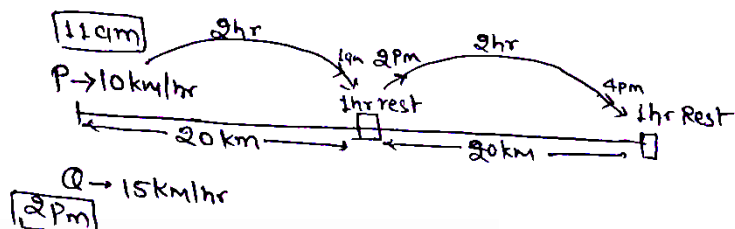
- 106** A man starts running from point P at 11:00am with speed of 10 km/hr. He runs for 2 hours and then takes a 1 hour rest. He continues this till he is caught by another man who starts from point P and runs non-stop at a speed of 15 km/hr towards

Soln:-

Covered distance by A
in 4 hr (11am - 4pm)

$$= 10 \times 4$$

$$= 40 \text{ km}$$



Covered distance by man B in 2 hr (2pm - 4pm)

$$= 15 \times 2 = 30 \text{ km}$$

Remaining distance = $(40 - 30) = 10 \text{ km}$

$$\text{Time} = \frac{10}{15} = \frac{2}{3} \times 60 = 40 \text{ min}$$

Time when man A caught by the man B = **4:40 pm** Ans

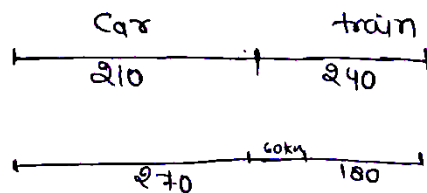
- 107** A man travels 450 km to his home partly by train and partly by car. He takes 8 hr 40 min if he travels 240 km by train and rest by car. he takes 20 mins more if he travels 180 km by train and the rest by car. The speed of the car in km/hr is:

Soln:- To travel extra 60 km by car
increase in time = 20 min

So travel extra 240 km by car
increase in time = 80 min

450 km by car in = 8 hr 40 min + 80 min = 10 hr

$$\text{Speed of car} = \frac{450}{10} = \boxed{45 \text{ km/hr}}$$



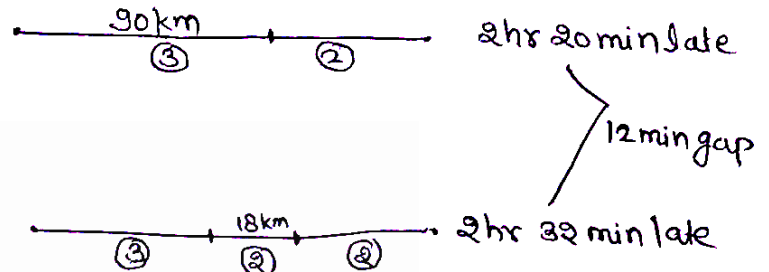
108 A train moves from a station and after travelling 90 km it meets with an accident. Due to this its speed reduce by $\frac{2}{3}$ rd and it reaches its destination 2 hour 20 min late. If his accident occurs 18 km before then it reaches to destination 2 hours 32 minutes late. Find the initial speed of the train and the distance between the station and its destination?

Soln:-

$$S \rightarrow 3 : 2$$

$$T \rightarrow 2 : 3$$

$$\begin{array}{c} \swarrow \quad \searrow \\ 24 \text{ min} \quad 12 \text{ min} \end{array}$$



$$\text{Speed} = \frac{18 \text{ km}}{\frac{24}{60}} = \boxed{45 \text{ km/hr}} \text{ Ans}$$

Distance:-

$$18 \text{ km before} \longrightarrow 12 \text{ min late}$$

$$8 \text{ km before} \longrightarrow 2 \text{ min late}$$

$$\downarrow \times 70$$

$$\downarrow \times 70$$

$$\underline{210 \text{ km}} \text{ before} \longrightarrow 140 \text{ min late (2hr 20min)} \\ \text{(To reach on time)}$$

$$\text{Distance} = 90 + 210$$

$$= \boxed{300 \text{ km}} \text{ Ans}$$

109 After travelling 25 km the speed of the car increase by $\frac{1}{4}^{th}$ of its original speed, due to this the car reaches 30 minutes earlier to its destination. If the speed of the car increased 10 km before, then it reaches to its destination $32\frac{2}{5}$ minutes earlier. Then find the distance travelled by car.

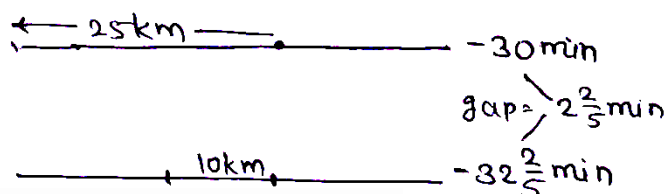
Soln:-

$$10 \text{ km} \longrightarrow 2\frac{2}{5} \text{ min}$$

$$5 \text{ km} \longrightarrow \frac{6}{5} \text{ min}$$

$$\downarrow \times 25 \quad \downarrow \times 25$$

$$125 \text{ km} \longrightarrow 30 \text{ min} \\ (\text{To reach on time})$$



$$\text{Distance} = 25 + 125 = \boxed{150 \text{ km}} \text{ Ans}$$

110 A, B and C can walk at the rates of 6, 8 and 10 km/hr respectively, they start from the same place at 2, 4 and 6 pm in the evening in the same direction respectively. When B catches A, B sends him back with a message to C. When will C get the message?

Soln

$$A \rightarrow 2 \text{ pm}, 6 \text{ km/hr}$$

$$B \rightarrow 4 \text{ pm}, 8 \text{ km/hr}$$

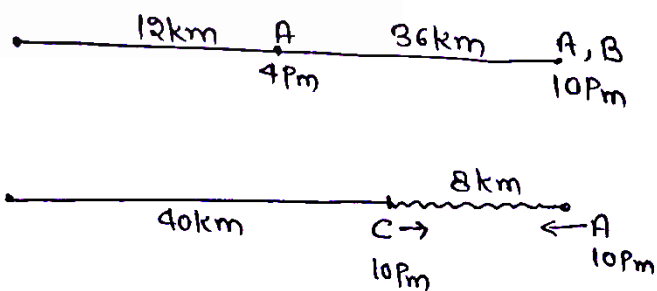
$$C \rightarrow 6 \text{ pm}, 10 \text{ km/hr}$$

$$B \text{ will catch } A = \frac{12}{8-6} = 6 \text{ hr}$$

$$A \text{ (in 6 hr)} = 6 \times 6 = 36 \text{ km}$$

$$\text{meeting time of C \& A} = \frac{8}{6+10} = \frac{1}{2} \text{ hr}$$

$$C \text{ get message} = \boxed{10:30 \text{ pm}} \text{ Ans}$$



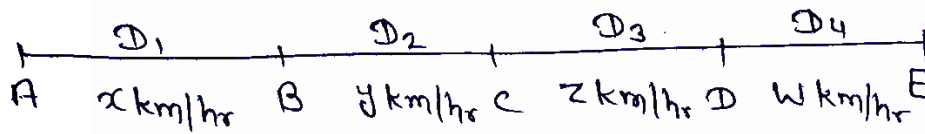
AVERAGE

SPEED

Average Speed

$$\text{Average speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

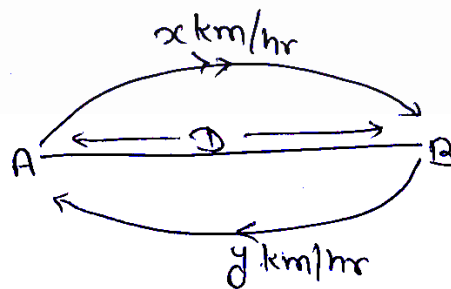
Concept 1:-



$$\text{avg. Speed} = \frac{D_1 + D_2 + D_3 + D_4}{\frac{D_1}{x} + \frac{D_2}{y} + \frac{D_3}{z} + \frac{D_4}{w}}$$

MathswithGajanand

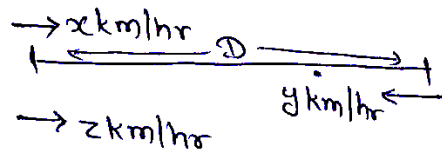
Concept - 2



$$\text{Avg. Speed} = \frac{D + D}{\frac{D}{x} + \frac{D}{y}} = \frac{2}{\frac{1}{x} + \frac{1}{y}}$$

$$\text{avg. Speed} = \frac{2xy}{x+y}$$

Concept-3



$$\text{avg. Speed} = \frac{3D}{\frac{D}{x} + \frac{D}{y} + \frac{D}{z}}$$

$$\text{avg. Speed} = \frac{3xyz}{xy + yz + zx}$$

- 1 A man travels 210 km with the speed of 60 km/hr and next 198 km with the speed of 66 km/hr. Find the avg speed of the whole journey.

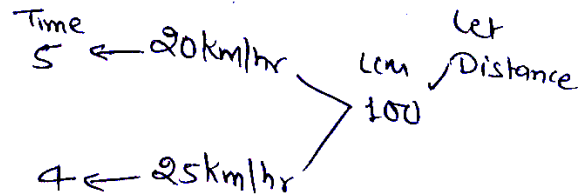
Soln:- avg. Speed, $= \frac{210 + 198}{\frac{210}{60} + \frac{198}{66}} = \boxed{62.8 \text{ km/hr}}$ Ans

- 2 A man walks from point X to Y at a speed of 20 km/h, but comes back from point Y to X at a speed of 25 km/hr. Find his average speed.

Soln:- mn-I

$$\begin{aligned} \text{avg. Speed} &= \frac{2xy}{x+y} \\ &= \frac{2 \times 20 \times 25}{(20 + 25)} \\ &= \frac{200}{9} \\ &= \boxed{22\frac{2}{9} \text{ km/hr}} \end{aligned}$$

mn-II



$$\begin{aligned} \text{avg. Speed} &= \frac{2 \times 100}{(5 + 4)} \\ &= \boxed{22\frac{2}{9} \text{ km/hr}} \text{ Ans} \end{aligned}$$

- 3] A car covers 15km, 20km, 30km and 12km at speeds 20 km/hr, 30 km/hr, 40 km/hr, and 30 km/hr, respectively. The average speed of the car for the total journey is?

Soln:-

$$\begin{aligned} \text{avg. Speed} &= \frac{(15+20+30+12)}{\left(\frac{15}{20} + \frac{20}{30} + \frac{30}{40} + \frac{12}{30}\right)} = \\ &= \frac{77}{\frac{3}{4} + \frac{4}{6} + \frac{3}{4} + \frac{2}{5}} \\ &= \frac{77}{\frac{77}{30}} = \boxed{30 \text{ km/hr}} \text{ Ans} \end{aligned}$$

- 4] Richa travels from A to B at the speed of 15 km/hr, from B to C at 20 km/h, and from C to D at 30 km/hr. If $AB = BC = CD$, then find the Richa's average speed.

Soln:- m-I

$$\begin{aligned} \text{avg. Speed} &= \frac{3xyz}{xy + yz + zx} \\ &= \frac{3 \times 15 \times 20 \times 30}{300 + 600 + 450} \\ &= \frac{3 \times 15 \times 20 \times 30}{1350} \\ &= \boxed{20 \text{ km/hr}} \text{ Ans} \end{aligned}$$

m-II

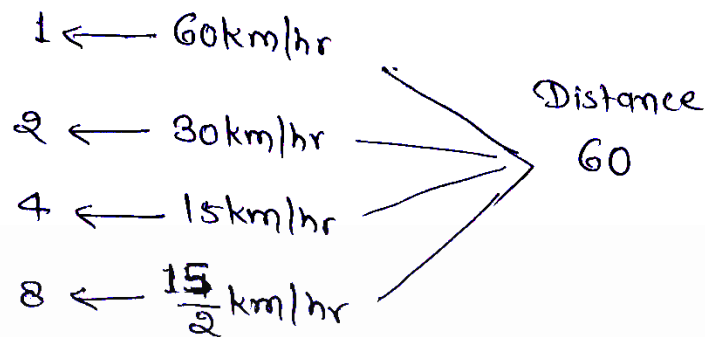
Time		
4	← 15 km/hr	$\begin{matrix} \text{LCM}(15, 20, 30) \\ \text{Distance} \end{matrix} \rightarrow 60$
3	← 20 km/hr	
2	← 30 km/hr	

$$\text{avg. Speed} = \frac{3 \times 60}{(4+3+2)}$$

$$= \boxed{20 \text{ km/hr}} \text{ Ans}$$

- 5] A man makes four trips of equal distance. His speed on first trip was 60 km/hr and in each subsequent trip his speed was half of the previous trip. What is the average speed (in km/hr) of the man in these four trips?

Soln:-



$$\begin{aligned} \text{avg. speed} &= \frac{4 \times 60}{(1+2+4+8)} = \frac{240}{15} \\ &= \boxed{16 \text{ km/hr}} \text{ Ans} \end{aligned}$$

- 6] A train travels the distance between stations P and Q at a speed of 126 km/h, while in the opposite direction it comes back at 90 km/hr. Another train travels the same distance at the average speed of the first train. The time taken by the second train to travel 525 km is?

Soln:-

$$\begin{aligned} \text{avg. speed} &= \frac{2 \times 90 \times 126}{216} \\ &= 105 \text{ km/hr} \end{aligned}$$

$$\begin{aligned} \text{Time} &= \frac{525}{105} \\ &= \boxed{5 \text{ hours}} \text{ Ans} \end{aligned}$$

- [7] A man goes to a place on bicycle at speed of 16 km/hr and comes back at lower speed. If the average speed is 6.4 km/hr in total, then the return speed (in km/hr) is:

Soln:-

$$\frac{6.4}{165} = \frac{2 \times 16 \times x}{(16+x)}$$

$$16+x = 5x$$

$$x = \boxed{4 \text{ km/hr}} \text{ Ans}$$

- [8] A boy walks to the park situated 500m from his house it took 5 min. and comes back in 7 min. What is the average speed of the boy (in km/hr)?

Soln

$$\begin{aligned} \text{avg. speed} &= \frac{500+500}{(5+7) \times 60} = \frac{1000}{12 \times 60} \text{ m/sec} \times \frac{18}{5} \\ &= \boxed{5 \text{ km/hr}} \text{ Ans} \end{aligned}$$

- [9] Shyam drives his car 30 km at a speed of 45 km/hr and for the next 1 hr 20 min, he drives it at a speed of 51 km/hr. Find his average speed (in km/hr) for the entire journey.

Soln:- Time = 1 hr 20 min = $\frac{4}{3}$ hr, speed = 51 km/hr

$$\text{Distance} = \frac{4}{3} \times 51 = 68$$

$$\begin{aligned} \text{avg. speed} &= \frac{30+68}{\frac{30}{45} + \frac{4}{3}} = \frac{98}{\frac{2}{3} + \frac{4}{3}} \\ &= \boxed{49 \text{ km/hr}} \text{ Ans} \end{aligned}$$

10 A man covers a total distance of 100km on bicycle. For the first 2 hours, the speed was 20km/hr and for the rest of the journey, it came down to 10 km/hr. The average speed will be!

Soln:- Distance in 1st 2hr = $20 \times 2 = 40\text{km}$
 Time taken to cover 60km = $\frac{60}{10} = 6\text{hr}$

$$\text{avg. speed} = \frac{100}{(2+6)} = \boxed{12\frac{1}{2} \text{ km/hr}} \text{ Ans}$$

11 One third of a certain journey is covered at the speed of 80km/hr one fourth of the journey at the speed of 50km/hr and the rest at the speed of 100km/hr. what will be the average speed (in km/hr) for the whole journey?

Soln:-

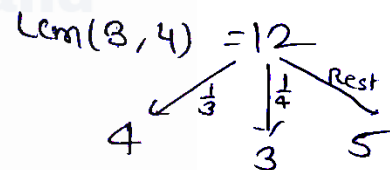
$$\text{avg Speed} = \frac{12}{\frac{4}{80} + \frac{3}{50} + \frac{5}{100}}$$

$$= \frac{12}{\frac{5 \times 20 + 6 + 5}{100}}$$

$$= \frac{12 \times 100}{816}$$

$$= \boxed{75 \text{ km/hr}} \text{ Ans}$$

Let Total Distance



[12] A person covers 40% of the distance from A to B at 8 km/hr, 40% of the remaining distance at 9 km/hr and the rest at 12 km/hr. His average speed (in km/hr) for the journey is:

Soln:- Let Total Distance = 100

```

    graph TD
      A[100] --> B[40]
      A --> C[24]
      A --> D[36]
      C --- E["60x40%"]
      D --- F["Remaining"]
  
```

$$\text{avg. speed} = \frac{100}{\frac{40}{8} + \frac{24}{9} + \frac{36}{12}} = \boxed{9\frac{3}{8} \text{ km/hr}} \text{ Ans}$$

[13] A Person travels a distance of 300m and then returns to the starting point. The time taken by him for the outward journey is 5 hours more than the time taken for the return journey. If he returns at a speed of 10 km/hr more than the average speed (in km/hr) for the entire journey?

Soln:- $\frac{300}{x} - \frac{300}{x+10} = 5 \text{ hr}$

$$60 \left[\frac{10}{x(x+10)} \right] = 1$$

$$600 = x(x+10)$$

Put $x = 20$

$$x = 20$$

$$\text{average Speed} = \frac{2 \times 20 \times 30}{(20 + 30)}$$

$$= \frac{2 \times 600}{50}$$

$$= \boxed{24 \text{ km/hr}} \text{ Ans}$$

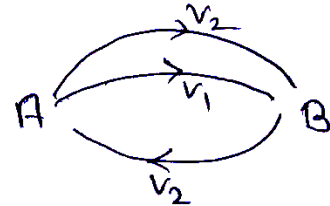
14 A car travels from A to B at V_1 km/hr, travels back from B to A at V_2 km/hr and again goes back from A to B at V_2 km/hr. The average speed of the car is:

Soln:-

$$\text{avg speed} = \frac{3V_1V_2V_2}{V_1V_2 + V_2^2 + V_1V_2}$$

$$= \frac{3V_1V_2V_2}{V_2(V_1 + V_2 + V_1)}$$

$$= \boxed{\frac{3V_1V_2}{2V_1 + V_2}} \quad \underline{\underline{\text{Ans}}}$$



15 Amit travelled from A to B at an average speed of 80 km/hr. He travelled the first 75% of the distance in two-third of the time and the rest at a constant speed of x km/hr. The value of x is:

Soln:-

Let total time = 3hr

$$\text{Total Distance} = 80 \times 3 = 240 \text{ km}$$

$$\begin{array}{|c|c|} \hline 75\% & 25\% \\ \hline 2\text{hr} & 1\text{hr} \\ \hline \end{array} \quad \rightarrow \quad 240 \times 25\% = 60 \text{ km}$$

$$x = \frac{60 \text{ km}}{1 \text{ hr}}$$

$$= \boxed{60 \text{ km/hr}} \quad \underline{\underline{\text{Ans}}}$$